DIMENSION OF A GROUPOID SCHEME OVER A FIELD WITH LOCALLY FINITE TYPE STRUCTURE MAPS EQUALS THE DIMENSION OF ITS STABILIZER GROUP SCHEME

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Allowed standard prerequisites (editorial).

- Uniqueness and geometric irreducibility of the irreducible component through the identity for a groupoid over a field (more-groupoids chapter, Lemma lemma-groupoid-on-field-geometrically-irreducible, canonical tag 04LT; collected on this lane as L22510). cited at proof lines 1562-1564. Provides the component Z through e, unique, thought of as an integral closed subscheme of R.
- The integral closure of the ground field in a function field of a variety is a field (varieties chapter, Lemma varieties-lemma-integral-closure-ground-field, canonical tag 04MI). cited at proof lines 1565-1568. Guarantees k'_s and k'_t are fields, the form in which the unique base field proposition applies.
- Uniqueness of the base field inside the function field of an integral variety (varieties chapter, Proposition varieties-proposition-unique-base-field, canonical tag 04MK). cited at proof lines 1569-1571. Gives k'_s = k'_t as subrings of Gamma(Z, O_Z), the pivot for s(k) = t(k).
- Equidimensionality of a groupoid over a field with locally finite type structure maps (more-groupoids chapter, Lemma lemma-groupoid-on-field-locally-finite-type-dimension, canonical tag 04LU; also in this lane's queue). cited at proof lines 1584-1586. Gives dim(R) = dim(Z), one side of the dimension squeeze.
- Equidimensionality of group schemes of finite type over a field (groupoids chapter, Lemma groupoids-lemma-group-scheme-finite-type-field, canonical tag 045X). cited at proof lines 1584-1586. Gives the equidimensionality of G, the other side of the squeeze.

Source reference. Original lemma 04LT: sources/stacks-chow/tags.txt, source lines 6019–6019. Cited at proof lines 1562-1564: uniqueness of the irreducible component Z through e.

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Source reference. Original lemma 04MI: sources/stacks-chow/tags.txt, source lines 6044–6044. Cited at proof lines 1565-1568: k'_s and k'_t are fields.

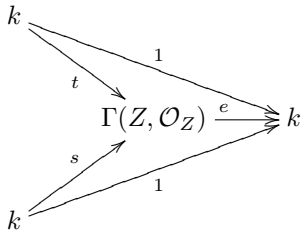
Source reference. Original proposition 04MK: sources/stacks-chow/tags.txt, source lines 6046–6046. Cited at proof lines 1569-1571: k'_s = k'_t as subrings of Gamma(Z, O_Z).

Source reference. Original lemma 04LU: sources/stacks-chow/tags.txt, source lines 6020–6020. Cited at proof lines 1584-1586: R is equidimensional, giving dim(R) = dim(Z).

Source reference. Original lemma 045X: sources/stacks-chow/tags.txt, source lines 5452–5452. Cited at proof lines 1584-1586: G is equidimensional, the other side of the squeeze.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(k)$ with k a field. Assume s, t are locally of finite type. Then $\dim(R) = \dim(G)$ where G is the stabilizer group scheme of R.*

Proof. Let $Z \subset R$ be the irreducible component passing through e (see Lemma 04LT) thought of as an integral closed subscheme of R. Let k'_s, resp. k'_t be the integral closure of $s(k)$, resp. $t(k)$ in $\Gamma(Z, \mathcal{O}_Z)$. Recall that k'_s and k'_t are fields, see Varieties, Lemma 04MI. By Varieties, Proposition 04MK we have $k'_s = k'_t$ as subrings of $\Gamma(Z, \mathcal{O}_Z)$. As e factors through Z we obtain a commutative diagram



This on the one hand shows that $k'_s = s(k)$, $k'_t = t(k)$, so $s(k) = t(k)$, which combined with the diagram above implies that $s = t$! In other words, we conclude that Z is a closed subscheme of $G = R \times_{(t,s),U \times_S U, \Delta} U$. The lemma follows as both G and R are equidimensional, see Lemma 04LU and Groupoids, Lemma 045X. $\square$